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Joint Optimization of Traffic Signal and Vehicle Route via Alternating Direction Method of Multipliers

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***Abstract*—This study proposes an integrated optimization model for simultaneous traffic signal and vehicle routing. The model jointly optimizes decentralized signal timing plans and individual vehicle routes to minimize vehicle delays. It explicitly captures the relationship between individual routing decisions and the traffic flow dynamics at signalized intersections. To enhance computational efficiency, an Alternating Direction Method of Multipliers (ADMM) is developed to decompose the integrated problem into several subproblems. Each vehicle can independently optimize its route while coordinating with the signal controller's decisions through information exchange. Simulation results demonstrate that the proposed method ensures better coordination between signal timing and vehicle routes, reducing the travel time and improving network efficiency. The ADMM algorithm solves each subproblem in approximately 0.1s, indicating the ability of real-time implementation in real-world scenarios.**

Keywords—Traffic Signal Control, Vehicle Routing, Connected Vehicle, ADMM.

I. Introduction

Traffic signal control and vehicle routing are promising approaches in urban traffic management to address congestion. These two approaches exist in a dynamic relationship of mutual influence that significantly impacts the overall performance of the road network[1]. Signal timing plans affect how vehicles choose their routes, while the route selection and turning rate of vehicles directly impact signal control decisions in return[2]. Therefore, it is necessary to consider the coordination of signal timing plans and vehicle route strategies to reduce congestion and improve traffic system efficiency.

Existing research on the coordination of signal optimization and route guidance can be divided into three main categories (i) dynamic routing response to signal plans, (ii) two-stage optimization framework, and (iii) joint integrated optimization. The first category focuses on developing dynamic routing strategies to accommodate fluctuating traffic conditions caused by signal control plans. For example, Dynamic System Optimal (DSO) and Dynamic User Equilibrium(DUE) strategies are widely used to adjust vehicle routes in response to the fluctuated link travel time caused by dynamic signal control plans[3, 4]. The DUE framework captures user-level rationality by modeling drivers' route choices under real-time conditions, whereas the DSO framework emphasizes system-wide efficiency by minimizing total network travel costs. However, these strategies respond passively to signal timing variations. They adapt vehicle routing decisions to prevailing traffic conditions but do not actively participate in optimizing the signal timing plans themselves. This disjointed adaptation constrains their effectiveness in delivering coordinated improvements across the entire transportation network.

The second category aims to coordinately optimize traffic signal and vehicle routing through the sequential decision-making structure[5, 6]. The coordination is typically modeled as a two-stage (or bi-level) problem. The first stage (or upper level) problem focuses on vehicle flow route assignment, while the second stage (or lower level) optimizes network signal plans based on the determined vehicle routes[7-9]. For example, Zaidi[10] proposed a framework that integrates DUO-based traffic assignment with adaptive back-pressure(BP) control. Vehicle flows are first allocated to routes to achieve user equilibrium, and the resulting traffic patterns are subsequently regulated by signal control. The iterative solution procedure is typically used to solve the two-stage problem. However, the two-stage framework sequentially optimizes signal timing and route guidance in an independent manner, making it difficult to guarantee a globally optimal solution.

The limitations in guaranteeing global optimality have motivated research on joint integrated optimization models, which aim to simultaneously optimize traffic signal plans and vehicle routing strategies[11]. Typically, such models are formulated as a centralized optimization problem of traffic flow assignment and network signal planning[12]. For example, De Souza[13] proposed a method to simultaneously optimize signal timing and vehicle routing based on a multi-commodity formulation of the store-and-forward model. Han[14] proposed a linear quadratic model predictive control approach to optimize signal timing and vehicle routing. However, the complexity of centralized models grows rapidly with network size. Few studies have proposed distributed algorithms to decompose the

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integrated problem into several sub-problems based on the augmented Lagrangian[15].

Existing integrated optimization models for traffic signal and vehicle routing still suffer from two main limitations. First, current research typically optimizes traffic flow assignment or turning rate. The aggregated traffic flow sacrifices the precise representation of each vehicle route as well as the flexibility of signal control. Second, the onboard capabilities at individual vehicles have not been fully utilized in existing studies. Many vehicles possess onboard computation resources through Connected Vehicle (CV) or advanced navigation technologies. Although several methods have proposed decomposition algorithms, they typically separate the problem into a large-scale route planning problem for all vehicles. It limits the scalability of integrated models in large-scale vehicle scenarios.

To address the research gap, this study formulates an integration optimization model for signal control and vehicle routing. The main contributions of this study are summarized as follows:

(1) We proposed an integrated model for joint optimization of decentralized signal control and individual vehicle routing. The proposed model enables simultaneous optimization by incorporating individual vehicle routing decisions into traffic flow dynamics.

(2) We designed an Alternating Direction Method of Multipliers (ADMM) algorithm. The problem is decomposed into sub-problems for single intersection signal optimization and individual vehicle route selection.

(3) The proposed method is applied to a real-world network, to evaluate its effectiveness and scalability in real-world scenarios.

The remainder of this study is organized as follows. The next section describes the integrated model of traffic signal and vehicle routing. In Section III, we propose the solution algorithm based on ADMM. Then, section IV demonstrates the effectiveness of the proposed model by extensive simulation experiments. Finally, we conclude this study in Section V.

II. Integrated Optimization Model

A. Problem Description

We consider an urban road network illustrated in Fig. 1. Each vehicle determines its movement direction when approaching an intersection. A typical isolated intersection provides three movement options on each approach: through, left-turn, and right-turn. It is assumed that all vehicles are CVs equipped with computational resources, namely CV controllers, which enable real-time route adjustments[16]. In addition, each signalized intersection is capable of dynamically adjusting its signal plans.

As shown in Fig. 1, each link is partitioned into two distinct zones: the route optimization zone and the no–route-changing zone. The route optimization zone corresponds to the upstream segment of the link, where vehicles are eligible for dynamic rerouting based on real-time traffic conditions. In this zone, vehicles continuously evaluate alternative routes and adjust their trajectories to improve overall efficiency. By contrast, the no–route-changing zone generally corresponds to the downstream segment of the link, particularly near intersections, where modifying routes is operationally infeasible because of safety constraints and lane-use restrictions[17-19]. Within this zone, vehicles are assumed to have committed to their current routes and are excluded from subsequent routing optimization.

B. Rolling Horizon Optimization Framework

To adapt the highly dynamic and uncertain nature of urban traffic, this study employs a rolling horizon optimization framework, which enables frequent updates of both signal plans and vehicle routes[20]. At each decision epoch, a new control strategy is formulated using real-time traffic information, ensuring that the framework adapts effectively to short-term fluctuations while maintaining robustness against longer-term variability. This adaptive framework improves the overall stability and efficiency of urban traffic management.

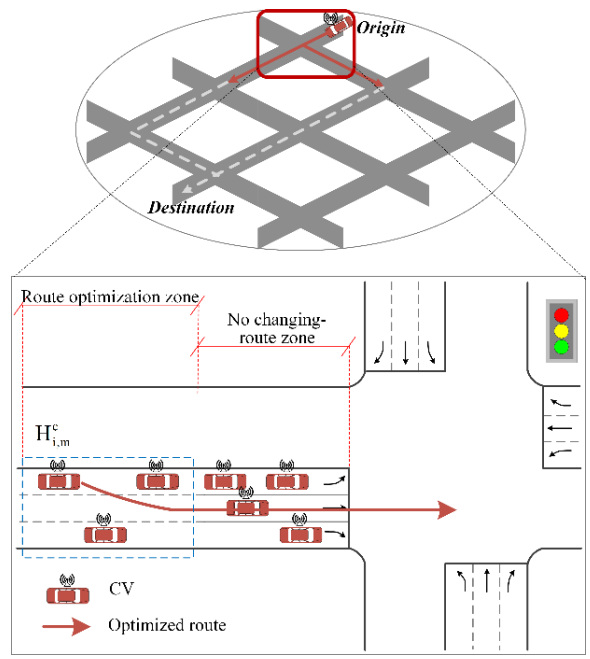

Figure 1. Typical isolated intersection within the network.

C. Decision Variables

We use p to denote the index of the optimization horizon. The decision variable of vehicle h is $\boldsymbol{z}_h = \varphi^k_{i,m,h,d}$, where $\varphi^k_{i,m,h,d}$ is a binary variable to represent vehicle h on movement m at intersection i whether choose the direction $d \in D$ in the kth optimization horizon. $\boldsymbol{z}_s = \{s^k_{i,m}, e^k_{i,m}\}_{m \in \mathrm{M}_i}$ is the variable of signal timing. $s^k_{i,m}, e^k_{i,m}$ denote the start and end times of the green interval for movement m in kth optimization horizon. M_i is the set of movements of intersection i. The set of all vehicles in the movement m at intersection i is denoted as $\mathrm{H}_{i,m}$. The set of all vehicles that in the route optimization zone is denoted as $\mathrm{H}^c_{i,m}$. $\mathrm{H}^c_{i,m}$ is a subset of $\mathrm{H}_{i,m}$. Only the vehicle $h \in \mathrm{H}^c_{i,m}$ can optimize its route in our study.

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D. Objective Function

The objective of the proposed model consists of two terms, where the first term is the intersection i's delay (U_i^k) and the second term is each CV h's travel cost (U_h^k). The formulation of U_i^k and U_h^k is shown in Eq. (1) and Eq. (2):

$$U_i^k = \sum_{m \in \mathrm{M}_i} q_{i,m}^{0,k} s_{i,m}^k + \pi \zeta_{i,m}^k \tag{1}$$

$$U_h^k = \sum_{m \in \mathrm{M}_i} \sum_{d \in D} \varphi_{i,m,h,d}^k \cdot T_{i,m,h,d}^k \tag{2}$$

where $q_{i,m}^{0,k}$ represents the initial queue length for movement m at intersection i during the kth optimization horizon; π is the correction factor; $\zeta_{i,m}^k$ is the residual queue vehicles of movement m carried over to the next optimization horizon. $T_{i,m,k,d}^k$ represents the travel cost of vehicle h on movement m at intersection i. The $T_{i,m,k,d}^k$ represents the shortest route cost to the destination of CV h when choosing direction d at movement m in intersection i, which is determined by the A* algorithm in this study[21]. If a direction d cannot reach the destination of CV h, the $T_{i,m,h,d}^k$ will be assigned a large number.

E. Model Constraints

a) Signal timing constraints $\mathbb{C}_S$

This section details the formulation of the signal timing plan constraints. The signal plan optimization problem is decentralized. Each intersection i only optimizes the local signal plan based on the neighbor intersections' outflow [22, 23].

In this study, traffic signal control is designed with a flexible and non-fixed phase structure, rather than relying on predetermined phase sequences[24]. Such a design enables the controller to dynamically adjust signal phases in response to real-time traffic conditions, thereby enhancing adaptability under fluctuating demands. To ensure operational safety, conflicting phases are strictly prohibited from being activated simultaneously, consistent with established principles in the signal control literature. This flexible phase mechanism allows the system to accommodate fluctuating traffic demands better while maintaining conflict-free operations at intersections.

For a certain movement m at intersection i, the green duration must lie within predefined minimum and maximum thresholds. For a certain movement m at intersection i, green duration should fall between the minimum and the maximum thresholds. Moreover, the green phase duration is also related to the number of queuing vehicles $q_{i,m}^{0,k}$, and the number of vehicles scheduled to depart $q_{i,m}^{out,k}$ [25].

$$e_{i,m}^k - s_{i,m}^k \geq \max\{g_{min}, q_{i,m}^{out,k} h_{avg} + q_{i,m}^{0,k} h_{min} + t_l\}, \quad m \in \mathrm{M}_i \tag{5}$$

$$e_{i,m}^k - s_{i,m}^k < \min\{g_{max}, q_{i,m}^{out,k} h_{avg} + q_{i,m}^{0,k} h_{min} + t_l\}, \quad m \in \mathrm{M}_i \tag{6}$$

where h_{avg}, h_{min} are the average and minimum headway between vehicles; t_l is the start-up lost time. g_{min}, g_{max} are the minimum and maximum green duration.

The dynamic evolution of queue lengths at signalized intersections, as follow:

$$\zeta_{i,m}^k = \max\{q_{i,m}^{0,k} + q_{i,m}^{in,k} - q_{i,m}^{out,k}, 0\}, m \in \mathrm{M}_i \tag{7}$$

$$q_{i,m}^{in,k} = \varpi_{i,m} \sum_{n \in \mathrm{U}_m} q_{i,n}^{out,k-1}, m \in \mathrm{M}_i \tag{8}$$

$$q_{i,m}^{out,k} = \sum_{h \in \mathrm{H}_{i,m}} \sum_{d \in D} \varphi_{i,m,h,d}^k \cdot \tag{9}$$

where $\zeta_{i,m}^k$ is the number of residual queue vehicles for movement m at intersection i in the optimization horizon k. $q_{i,m}^{in,k}$ is the number of arrival vehicles during the optimization horizon k; U_m denotes the set of upstream movements of m; $q_{i,m}^{out,k}$ is the number of vehicles scheduled to depart in the pth optimization horizon, which is determined by variable $\varphi_{i,m,h,d}^k$. $\varpi_{i,m}$ is turning rate for movement m at intersection i.

This study adopts a rolling horizon framework. Therefore, the signal plan may be updated frequently. To ensure a smooth transition between two adjacent updated plans, Eq.(10) and (11) should be introduced[26].

$$e_{i,m}^k = e_{i,m}^{k-1} | (s_{i,m}^{k-1} \leq t_p \ and \ e_{i,m}^{k-1} > t_p), m \in \mathrm{M}_i \tag{10}$$

$$s_{i,m}^k \geq e_{i,m}^{k-1} + \psi - M(1 - \gamma_{m,n}) | (e_{i,m}^{k-1} > t_p), m \in \mathrm{M}_i \tag{11}$$

b) Routing constraints $\mathbb{C}_R$

At an intersection, vehicles selecting different movements will encounter varying route lengths. If a candidate movement does not provide a route to the vehicle's destination, the corresponding indicator variable $T_{i,m,k,d}^k$ is assigned an infinite weight. This prevents the vehicle from selecting an infeasible movement. Conversely, for feasible movements, a bounded weight is assigned that represents the shortest route distance to the destination.[11] We apply the A* algorithm to determine the shortest route and compute the route length from the origin node to the destination node.

$$T_{i,m,h,d}^k = \begin{cases} L_{route} & R_{i,m,d} \neq \emptyset \\ +\infty & R_{i,m,d} = \emptyset \end{cases}, m \in \mathrm{M}_i, h \in \mathrm{H}_{i,m}^c, d \in D \tag{12}$$

where $R_{i,m,d}$ denotes the destination movement of vehicle h, defined as the downstream link of the corresponding turning direction. The set comprises all feasible routes from movement m to $R_{i,m,d}$, the destination movement of vehicle h. If non-empty, its value is the minimum route length from the current position to the destination; otherwise, the movement is assigned an infinite route length.

A fundamental constraint in discrete vehicle routing is that each vehicle must be assigned to one, and only one, route from its set of feasible routes. This principle stems from the physical

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reality that a vehicle h is an indivisible entity and cannot simultaneously traverse multiple routes. To mathematically enforce this single-route assignment, we introduce the binary choice constraint presented in Eq. (13):

$$\sum_{d\in \mathrm{D}} \varphi_{i,m,h,d}^{k} = 1, m \in \mathrm{M}_i, h \in \mathrm{H}_{i,m}^{\mathrm{c}} \tag{13}$$

F. Model Overall Formulation

In summary, we formulate the overall formulation of the integrated optimization model as Eq. (14):

$$\min J^k = \alpha U_i^k(\boldsymbol{z}_h, \boldsymbol{z}_{\mathrm{s}}) + \sum_{m\in \mathrm{M}_i} \sum_{h\in \mathrm{H}_{i,m}^{\mathrm{ctrl}}} \beta U_h^k(\boldsymbol{z}_h) \tag{14}$$

subject to:

$$(\boldsymbol{z}_h, \boldsymbol{z}_{\mathrm{s}}) \in \mathbb{G}_{\mathrm{S}}; (\boldsymbol{z}_h) \in \mathbb{G}_{\mathrm{R}}$$

where α and β are weighting factors to balance the objectives of signal optimization and vehicle routing.

III. ADMM Algorithm

The optimization problem formulated in Eq. (14) has a key structural property: the objective function is separable with respect to the decision variables of each CV and signal controller. This separability makes the problem amenable to decomposition[27, 28]. To exploit this structure for computational efficiency, we develop a distributed solution framework based on the ADMM. The ADMM algorithm solves large-scale optimization problems by decomposing them into subproblems, which are then solved iteratively and in parallel for each CV and each signal controller. Local solutions are coordinated via dual-variable updates to drive convergence toward a global optimum.

A. Reformulation of the Consensus Problem

We use $\mathbb{V}_h$ to denote the CV controller. Each intersection i is also equipped with a signal controller, which is denoted as $\mathbb{S}_i$. The signal controller $\mathbb{S}_i$ needs to optimize $\boldsymbol{z}_h^{\mathbb{S}_i}$ and $\boldsymbol{z}_{\mathrm{s}}^{\mathbb{S}_i}$. The CV controller is associated with the optimization of $\boldsymbol{z}_h^{\mathbb{V}_h}$.

To formally capture the feasibility requirements of both the signal timing plan and vehicle routing, we introduce two indicator functions $\mathbb{I}_{\mathrm{S}}(\cdot)$ and $\mathbb{I}_{\mathrm{R}}(\cdot)$ to describe constraints of signal timing plan and vehicle route. By incorporating these indicator functions into the formulation, the original problem can be equivalently expressed as a consensus optimization framework. The problem (15) could be rewritten as follow:

$$\min \hat{J}^k = \underbrace{\left[\alpha U_i^k\left(\boldsymbol{z}_h^{\mathbb{S}_i}, \boldsymbol{z}_{\mathrm{s}}^{\mathbb{S}_i}\right) + \mathbb{I}_{\mathrm{S}}\left(\boldsymbol{z}_h^{\mathbb{S}_i}, \boldsymbol{z}_{\mathrm{s}}^{\mathbb{S}_i}\right)\right]}_{\text{objective of signal controller}} + \underbrace{\sum_{m\in \mathrm{M}_i} \sum_{h\in \mathrm{H}_{i,m}^{\mathrm{c}}} \left[\beta U_h^k\left(\boldsymbol{z}_h^{\mathbb{V}_h}\right) + \mathbb{I}_{\mathrm{R}}\left(\boldsymbol{z}_h^{\mathbb{V}_h}\right)\right]}_{\text{objective of CV controller}} \tag{15}$$

subject to:

$$\boldsymbol{z}_h^{\mathbb{S}_i} = \boldsymbol{c}_h, \boldsymbol{z}_h^{\mathbb{V}_h} = \boldsymbol{c}_h, \forall h \in \mathrm{H}_{i,m}^{\mathrm{c}}$$

where $\boldsymbol{c}_h$ is the consensus variable of vehicle route selection.

B. Distributed Algorithm

We used the ADMM algorithm to decompose the consensus problem in Eq. (16) into multiple subproblems.

$$\begin{aligned} L_\rho = &\left[\alpha U_i^k\left(\boldsymbol{z}_h^{\mathbb{S}_i}, \boldsymbol{z}_{\mathrm{s}}^{\mathbb{S}_i}\right) + \mathbb{I}_{\mathrm{S}}\left(\boldsymbol{z}_h^{\mathbb{S}_i}, \boldsymbol{z}_{\mathrm{s}}^{\mathbb{S}_i}\right)\right] \\ &+ \sum_{h\in \mathrm{H}_{i,m}^{\mathrm{c}}} \frac{\rho}{2}\left\|\boldsymbol{z}_h^{\mathbb{S}_i} - \boldsymbol{c}_h + \boldsymbol{\lambda}_h^{\mathbb{S}_i}\right\|^2 \\ &+ \sum_{m\in \mathrm{M}_i} \sum_{h\in \mathrm{H}_{i,m}^{\mathrm{c}}} \left[\beta U_h^k\left(\boldsymbol{z}_h^{\mathbb{V}_h}\right) + \mathbb{I}_{\mathrm{R}}\left(\boldsymbol{z}_h^{\mathbb{V}_h}\right)\right] \\ &+ \sum_{h\in \mathrm{H}_{i,m}^{\mathrm{c}}} \frac{\rho}{2}\left\|\boldsymbol{z}_h^{\mathbb{V}_h} - \boldsymbol{c}_h + \boldsymbol{\lambda}_h^{\mathbb{V}_h}\right\|^2 \end{aligned} \tag{16}$$

where $\boldsymbol{\lambda} = \left[\boldsymbol{\lambda}_h^{\mathbb{S}_i}, \boldsymbol{\lambda}_h^{\mathbb{V}_h}\right]^{\mathrm{T}}$ is the scaled dual variable, and ρ is the penalty parameter.

We use μ to denote the index of iteration in the ADMM. The ADMM framework addresses the augmented Lagrangian function (16) through an iterative process consisting of three steps. This structured approach enables the decomposition of the original optimization problem into smaller and more tractable subproblems, which are coordinated through consensus updates to ensure convergence.

Step1: The signal controller $\mathbb{S}_i$ updates control variables $\boldsymbol{z}_h^{\mathbb{S}_i,\mu+1}$ and $\boldsymbol{z}_{\mathrm{s}}^{\mathbb{S}_i,\mu+1}$, Each CV controller updates control variables $\boldsymbol{z}_h^{\mathbb{V}_h,\mu+1}$:

$$\begin{aligned} \boldsymbol{z}_h^{\mathbb{S}_i,\mu+1}, \boldsymbol{z}_{\mathrm{s}}^{\mathbb{S}_i,\mu+1} = \operatorname{argmin} \Bigg\{ &\left[\alpha U_i^k\left(\boldsymbol{z}_h^{\mathbb{S}_i}, \boldsymbol{z}_{\mathrm{s}}^{\mathbb{S}_i}\right) + \mathbb{I}_{\mathrm{S}}\left(\boldsymbol{z}_h^{\mathbb{S}_i}, \boldsymbol{z}_{\mathrm{s}}^{\mathbb{S}_i}\right)\right] \\ &+ \sum_{h\in \mathrm{H}_{i,m}^{\mathrm{c}}} \frac{\rho}{2}\left\|\boldsymbol{z}_h^{\mathbb{S}_i} - \boldsymbol{c}_h^{\mu} + \boldsymbol{z}_h^{\mathbb{S}_i,\mu}\right\|^2 \Bigg\} \end{aligned} \tag{17}$$

Each CV controller solves the following subproblem in parallel to update control variables $\boldsymbol{z}_h^{\mathbb{V}_h,\mu+1}$:

$$\begin{aligned} \boldsymbol{z}_h^{\mathbb{V}_h,\mu+1} = \operatorname{argmin} \Big\{ &\left[\beta U_h^k\left(\boldsymbol{z}_h^{\mathbb{V}_h}\right) + \mathbb{I}_{\mathrm{R}}\left(\boldsymbol{z}_h^{\mathbb{V}_h}\right)\right] \\ &+ \frac{\rho}{2}\left\|\boldsymbol{z}_h^{\mathbb{V}_h} - \boldsymbol{c}_h^{\mu} + \boldsymbol{\lambda}_h^{\mathbb{V}_h,\mu}\right\|^2 \Big\} \end{aligned} \tag{18}$$

The convex quadratic programming problem in (17)-(18) could be solved with existing optimization algorithms[29].

Step2: Updating $\boldsymbol{c}_h$ during each iteration enables the algorithm to progressively reconcile discrepancies among subproblems, which guarantees convergence toward a globally feasible and optimal solution. The relevant formulas are as follows:

$$\boldsymbol{c}_h^{\mu+1} = \frac{\left(\boldsymbol{c}_h^{\mathbb{V}_h,\mu+1} + \boldsymbol{\lambda}_h^{\mathbb{V}_h,\mu}\right) + \left(\boldsymbol{c}_h^{\mathbb{S}_i,\mu+1} + \boldsymbol{\lambda}_h^{\mathbb{S}_i,\mu}\right)}{1+1}, \forall h \in \mathrm{H}_{i,m}^{\mathrm{c}} \tag{19}$$

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Step3: The scaled dual variable λ is updated in each controller as (20)-(21):

$$\boldsymbol{\lambda}_h^{\mathbb{V}_h,\mu+1} = \boldsymbol{\lambda}_h^{\mathbb{V}_h,\mu} + \left(\boldsymbol{z}_h^{\mathbb{V}_h,\mu+1} - \boldsymbol{c}_h^{\mu+1}\right), \forall h \in \mathrm{H}_{i,m}^{\mathrm{c}} \quad (20)$$

$$\boldsymbol{\lambda}_h^{\mathbb{S}_i,\mu+1} = \boldsymbol{\lambda}_h^{\mathbb{S}_i,\mu} + \left(\boldsymbol{z}_h^{\mathbb{S}_i,\mu+1} - \boldsymbol{c}_{\boldsymbol{h}}^{\mu+1}\right), \forall k \in \mathrm{H}_{i,m}^{\mathrm{c}} \quad (21)$$

The stopping criterion is defined such that the iterative process terminates once the error metric falls below a predetermined threshold, ensuring convergence of the ADMM algorithm.

IV. Evaluation

This section evaluates the proposed integrated optimization model for signal planning and vehicle routing within a real-world road network. The evaluation comprises five components: experimental design, benchmark, global mobility, sensitivity, and computational efficiency.

A. Experiment Design

The performance of the proposed model was evaluated using a real-world road network from Qinzhou, China, as shown in Fig. 2. Model validation was conducted through experiments in the SUMO platform[30]. The simulation is controlled through the interface TraCI in Python 3.11 with Gurobi as a solver. The delay in communications is assumed to be negligible. Each simulation runs for 1800 s, including a 300 s warm-up phase.

The traffic demand data was collected on December 20, 2023. It's used to evaluate the effectiveness of the proposed model, except in the sensitivity analysis conducted under varying demand levels.

B. Benchmarks

To evaluate the performance and effectiveness of the proposed model, we compare it against several representative benchmark strategies.:

- **FSR (Fixed Signal and Route)**: This method maintains the original signal timing plan and vehicle routes without any optimization. It reflects existing traffic conditions and serves as a baseline for comparison.
- **DUE (Dynamic User Equilibrium):** This models a dynamic routing process under user equilibrium conditions. Each vehicle updates its route choice based on link travel times. At equilibrium, no individual vehicle can further reduce its travel time by unilaterally switching to another route, thereby reflecting user-level rationality in routing decisions.
- **DUE-BP**[31]: This method represents a traditional two-stage framework. In the first stage, vehicle routes are determined by the DUE strategy. In the second stage, the BP algorithm is applied to optimize the signal timing plan based on the updated turning movements generated by DUE. This sequential structure optimizes routing and signal control independently rather than jointly.
- **Proposed**: The proposed integrated model optimizes decentralized signal plans and individual vehicle routes at the same time.

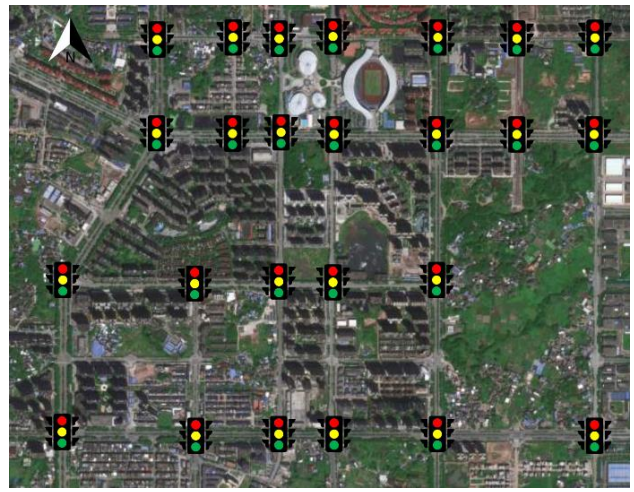

Figure 2. The real-world road network of Qinzhou City

C. Global Mobility

Table I shows statistics of CV travel times under the benchmark. The proposed model consistently outperforms all benchmarks. The proposed model attains the lowest mean travel time, 277.12s, corresponding to reductions ranging from 5.78% to 28.80% relative to the benchmark strategies. These results demonstrate substantial efficiency gains from jointly optimizing signal control and vehicle routing.

Beyond the mean, its standard deviation is moderate compared with most models, indicating more stable travel-time performance. The 95% confidence interval for the proposed model is 261.25s~282.46s, which is narrower and shifted toward shorter travel times than those of the benchmarks, indicating higher reliability. The median travel time further confirms that most vehicles experience consistent improvements rather than improvements only in the mean.

TABLE I. Comparison of the travel time of CVs

	Mean	SD	Lower 95%	Upper 95%	Median
FSR	389.18	119.88	378.06	400.30	416
DUE	337.45	128.54	321.78	353.12	364
DUE-BP	294.49	83.58	286.88	302.11	303
Proposed	277.12	86.85	261.25	282.46	284

Table 2 shows the comparison of rerouting behavior in DUE, DUE-BP, and the proposed method, using the FSR method as the baseline. It only counts the rerouted vehicle results compared to the FSR method. The proposed method exhibits the highest rerouting frequency (1.81) and probability (65.81%), indicating that more vehicles adjust their routes in response to current traffic conditions. Importantly, it achieves the largest travel time reduction for rerouted vehicles (-91.36 s), demonstrating the effectiveness of integrating signal control and routing decisions to alleviate congestion and improve efficiency.

In comparison, the DUE-BP method leads to more rerouting actions than DUE, but its average time savings (-33.71 s) are

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significantly less than those of the proposed model. This suggests that although DUE-BP increases responsiveness, it lacks effective coordination with the signal, limiting its benefit.

DUE shows the lowest rerouting rate and even increases the travel time of rerouted vehicles by 15.63 s on average. Although DUE initially reroutes vehicles to less congested links, it causes many vehicles to follow the same shortest routes, which later become heavily congested. As a result, rerouted vehicles actually experience longer delays. Meanwhile, other vehicles that are not rerouted may benefit from reduced congestion elsewhere, leading to a lower total system travel time at the expense of rerouted vehicles' efficiency.

These findings demonstrate that more frequent rerouting alone is not sufficient. It must be paired with coordinated signal optimization to effectively reduce travel times.

TABLE II. COMPARISON OF REROUTING BEHAVIOR IN DUE, DUE-BP, AND PROPOSED METHODS.

	Rerouted frequency	Rerouted probability	Reduced travel time of rerouted vehicles compared to FSR /s
DUE	0.70	50.30%	+15.63
DUE-BP	1.56	60.10%	-33.71
Proposed	1.81	65.81%	-91.36

D. Sentisitive

We evaluate the proposed model under five traffic demand levels spanning from under-saturated to over-saturated conditions: low, medium, high, saturated, and over-saturated. Figure 3 presents a comparison of the average travel times for the proposed model and benchmark methods across all demand levels. Average travel times increase for all methods as traffic demand rises; however, the proposed model consistently achieves significantly lower travel times than all benchmarks. Under saturated and over-saturated conditions, the proposed model effectively limits congestion propagation, enhances overall network throughput, and produces only modest increases in travel time. These findings demonstrate that the proposed model maintains robust performance and substantial efficiency gains across a wide range of traffic demand levels.

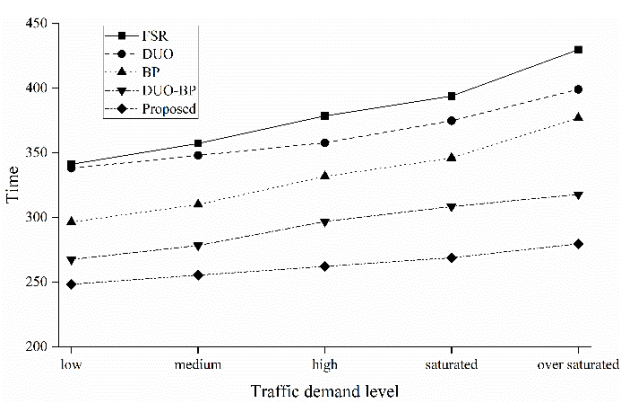

Figure 3. Travel time under various demand levels

E. Computational Efficiency

In this section, the centralized solution is employed as the benchmark, which directly solves problem (9) in a centralized fashion. Table I reports the average and maximum computation times for the two methods. The results highlight the substantial computational advantage of the ADMM-based algorithm over the centralized approach. Specifically, the ADMM algorithm decomposes the large-scale optimization problem into smaller subproblems that can be solved independently by each controller. This distributed structure not only significantly reduces computation time but also enhances scalability, allowing the framework to handle larger urban networks with many intersections and vehicles. Furthermore, the fast subproblem-solving capability, on the order of milliseconds, ensures that the algorithm can provide near real-time signal control and route guidance, making it highly suitable for practical deployment in dynamic traffic environments.

TABLE III. COMPARISON OF ALGORITHM SOLVING PERFORMANCE

Solution Type	Controller	Average computation time/s	Maximum computation time/s
ADMM algorithm	Vehicle	0.036	0.041
	Signal	0.124	0.279
Centralized	--	1.878	3.071

Fig.4 illustrates the convergence performance of the ADMM algorithm. It shows the primal residuals of the control variables for one CV controller during one optimization in the high-demand condition. The decreasing trend in primal residuals indicates that the local and consensus variables are progressively aligning. As the number of iterations increases, the residuals stabilize and diminish, reflecting convergence toward the optimal solution. In this case, convergence is achieved at the 11th iteration. Across our experiments, even under saturated or oversaturated traffic conditions, the optimization typically converges within approximately 10 iterations, demonstrating the algorithm's efficiency and robustness.

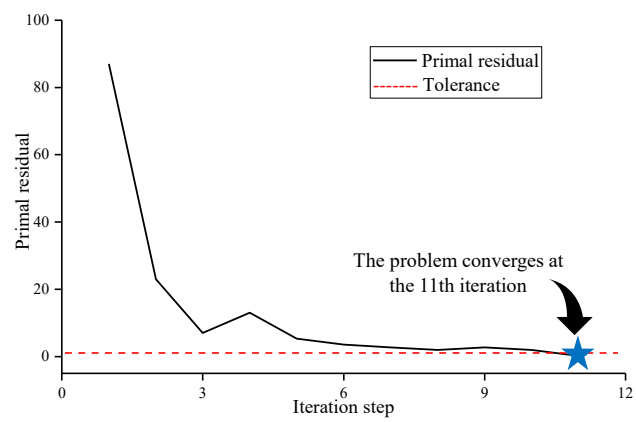

Figure 4. Convergence performance of the ADMM algorithm for one CV controller during one optimization

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V. CONCLUSION

This study proposes an integrated signal control and vehicle routing model that simultaneously optimizes signal plans at intersections and route choices for individual vehicles. By embedding individual routing decisions into the traffic flow dynamics, the model enables coordinated optimization and effectively captures the interactions between routing and signal control. To address the computational challenges of large-scale scenarios, ADMM algorithm is developed. Evaluation on a real-world urban road network demonstrates the effectiveness of the proposed model. The results show that:

- The proposed model reduces overall travel time significantly and enhances overall traffic efficiency.
- The ADMM algorithm can ensure each subproblem is solved in approximately 0.1 s, indicating its feasibility for real-world implementation.

Future research may explore several promising directions. First, the proposed model should be evaluated in the mixed traffic scenario with both regular vehicles and CVs. Further studies are needed to evaluate model performance under varying CV penetration and compliance rates. Second, the robustness of the framework can be improved by accounting for uncertainties in demand, vehicle behavior, and communication delay.

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